

5. A hexagonal plane of 35 mm side, has its one edge on H.P., which is 30° inclined to V.P. Draw the projections of hexagonal plane, when its surface is 45° inclined to H.P.

Solution : Refer fig. 6.31. Given $\theta = 45^\circ$.

Step (i) :

- 1) Since an edge of plane is on H.P., and surface of plane is $\angle$ to H.P.; first assume that plane is lying on H.P. Hence, draw T.S. (pentagon) on H.P. (in T.V.) with a side (a-b) at left and $\perp$ to x-y line.

- 2) From this T.V. of plane, project F.V. of plane, which is a surface line $a'b'-c'f'-d'e'$ on x-y line.

Step (ii) : Now, tilt the plane and make it (surface line $a'b'-c'f'-d'e'$ in F.V.) $45^\circ \angle$ to H.P. (x-y line). Project its T.V. $a_1b_1c_1d_1e_1f_1$ from this F.V. and T.V. of step (i).

Step (iii) : Since side on H.P. is $30^\circ \angle$ to V.P., reproduce the T.V. $a_1b_1c_1d_1e_1f_1$ with that side (a_1-b_1) $30^\circ \angle$ to x-y. Project final F.V. $a'_1b'_1c'_1d'_1e'_1f'_1$ from this T.V. and F.V. of step (ii).

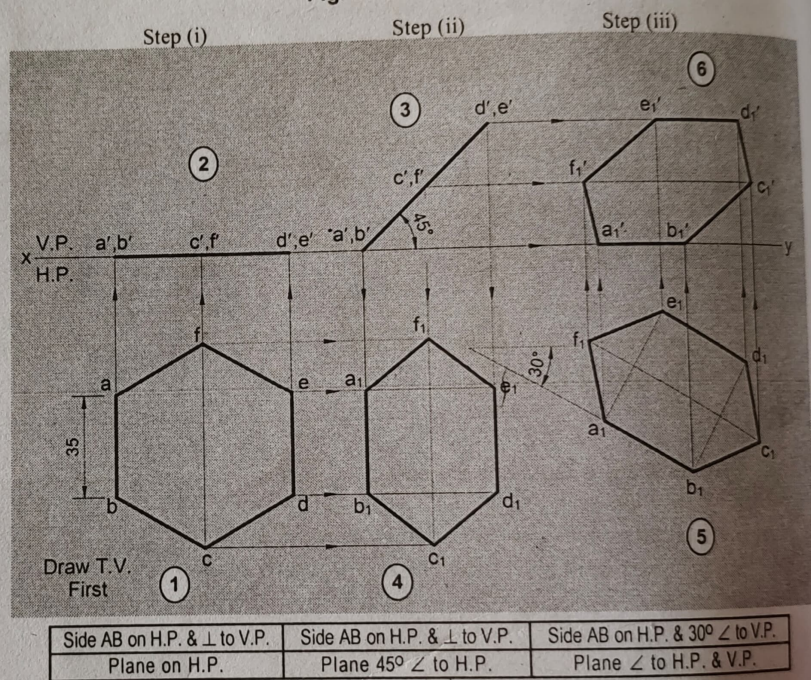


Fig. 6.31

17. A square ABCD of side length 40 mm is resting on its corner A on H.P. and diagonal BD is $\perp$ to V.P. in step (i).

H.P. and diagonal BD is 30° inclined to V.P. Draw its projections, when its surface is 50° inclined to

Solution : Refer fig. 6.43.

Procedure :

Step (i) : Since corner A is on H.P., and plane is $\angle$ to H.P.; first assume that complete plane is on H.P. We know that, when plane rests on H.P. completely (or $\parallel$ to H.P.), its T.V. shows its True Shape (T.S.).

1) Hence, on H.P. (in T.V.), draw T.S. of plane, which is a square abcd of side 40 mm. Keep corner a at left and diagonal b-d $\perp$ to x-y.

2) From this T.V. of plane, project its F.V., which is a surface line $a'-b'd'-c'$ on x-y line.

Step (ii) : Now, keeping the corner A on H.P., make the plane 50° $\angle$ to H.P.

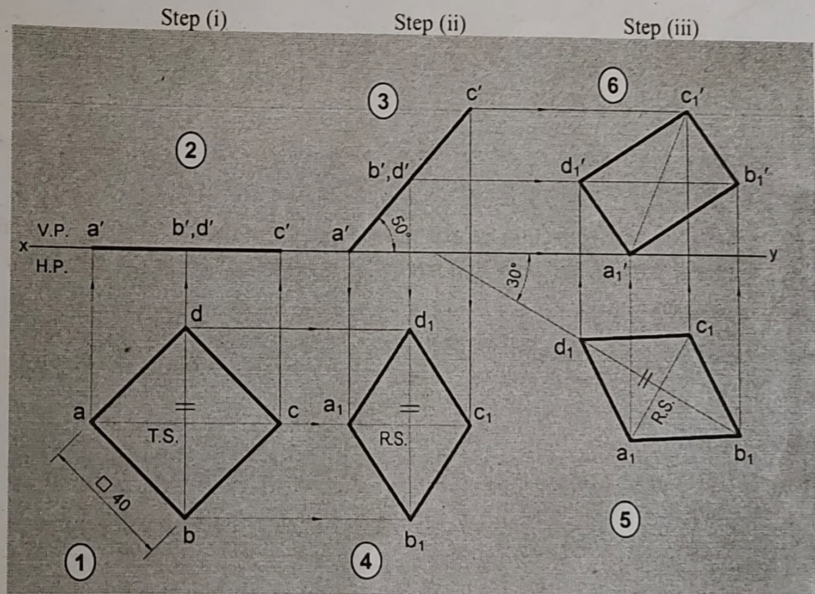
3) Hence, in F.V., mark a' on x-y line and redraw surface line $a'-b'd'-c'$ 50° $\angle$ to x-y line.

4) Draw vertical projection lines from each corner of $a'-b'd'-c'$ and also draw horizontal projection lines from T.V. of step (i) and obtain new T.V. $a_1b_1c_1d_1$, which is the reduced shape (R.S.) of plane. [Note that, in T.V., length of diagonal a_1-c_1 reduces, but length of diagonal b_1-d_1 remains same].

Step (iii) : Make the diagonal BD 30° $\angle$ to V.P.

5) Since the diagonal BD is 30° $\angle$ to V.P., reproduce the T.V. $a_1b_1c_1d_1$ with that diagonal b_1-d_1 30° $\angle$ to x-y line.

6) Project final F.V. $a_1'b_1'c_1'd_1'$ from this T.V. and F.V. of step (ii) as shown.



Diagonal BD $\perp$ to V.P.	Diagonal BD $\perp$ to V.P.	Diagonal BD 30° $\angle$ to V.P.
Plane on H.P.	Plane 50° $\angle$ to H.P.	Plane $\angle$ to H.P. & V.P.

Fig. 6.43

Note : When a diagonal of plane is $\angle$ to V.P. and $\parallel$ to H.P., then in step (i) it is $\perp$ to V.P. Like in this problem, in step (iii) Diagonal BD is $\angle$ to V.P. and $\parallel$ to H.P., which is $\perp$ to V.P. in step (i).

Fig. 6.101

A.2. A pentagonal plane of 30 mm side has its side on V.P. and 45° inclined to H.P. Draw its projections when the plane is 30° inclined to V.P.

Solution : Refer fig. 6.102.

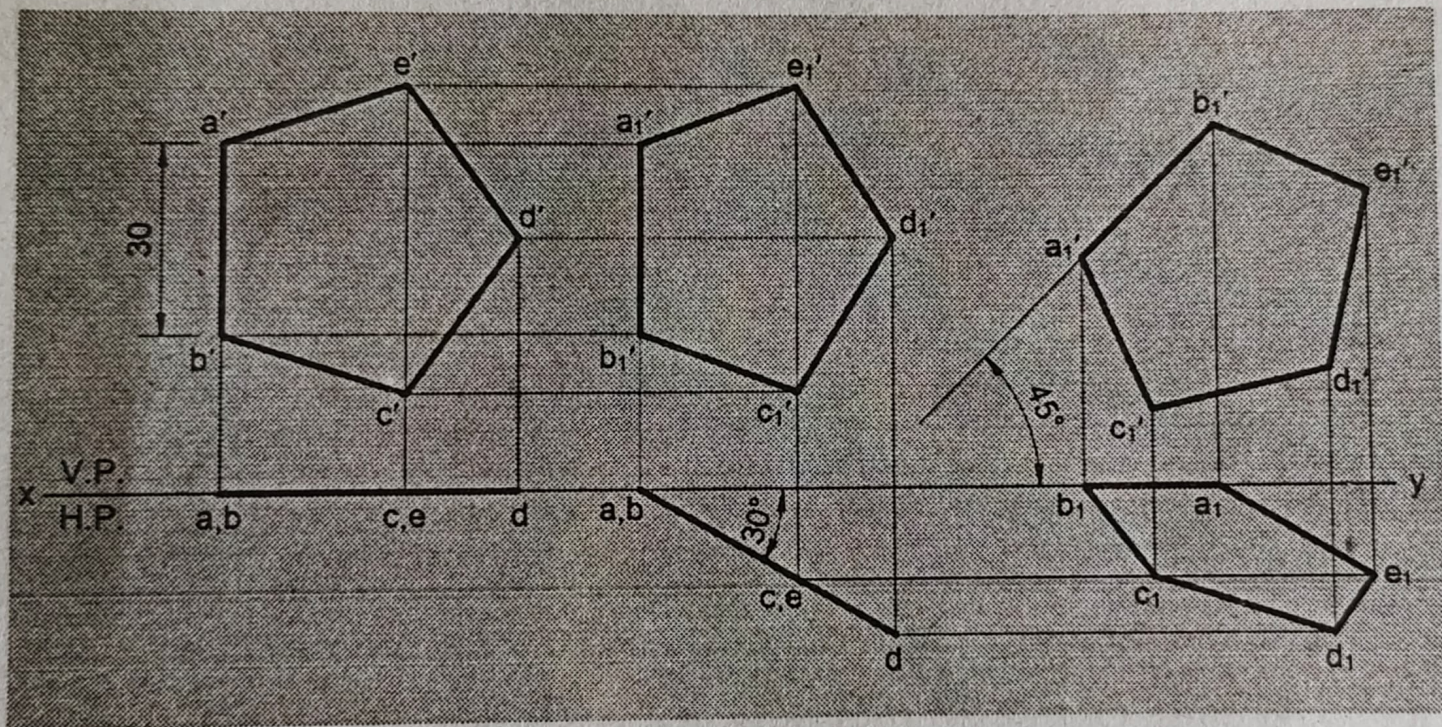


Fig. 6.102

42. Draw the projections of a lamina 50 mm diameter, resting in H.P. on a point 'A' on the circumference. Draw the projections of lamina with its plane inclined at 45° to H.P. and

- the top view of diameter AB making 30° angle with V.P.
- diameter AB making angle 30° with V.P.

Solution : Refer fig. 6.68.

Step (i) : In T.V., draw circle. Divide it into equal parts. Project its F.V.

Step (ii) : In F.V., make the plane 45° $\angle$ to x-y line. Project its T.V.

Step (iii) (a) : It is required to make T.V. of diameter AB 30° $\angle$ to V.P. In T.V., a_1-b_1 is the T.V. of diameter AB. Hence in step (iii) reproduce the T.V. with a_1-b_1 30° $\angle$ to x-y line. Then, project the F.V.

Step (iii) (b) : It is required to make diameter AB itself 30° $\angle$ to V.P. But, T.V. of AB i.e. a_1-b_1 is the Reduced Length of AB. Hence a_1-b_1 will make an angle (β) greater than 30° with x-y. This apparent angle (β) can be determined as described below.

In T.V., mark a_1 (any where). Draw line a_1-b_2 ($= a-b = T.L. = 50$ mm) 30° $\angle$ to x-y line. Draw locus of b through b_2 . With center a_1 and radius $= a_1-b_1$ [from Step (ii)] cut the locus of b at b_1 . Join a_1-b_1 . With a_1 and b_1 complete the T.V. Draw final F.V. from this T.V. and F.V. of step (ii).

12 Marks, Dr.B.A.T.U. [May 2018]

Dr.B.A.T.U. [May 2017]

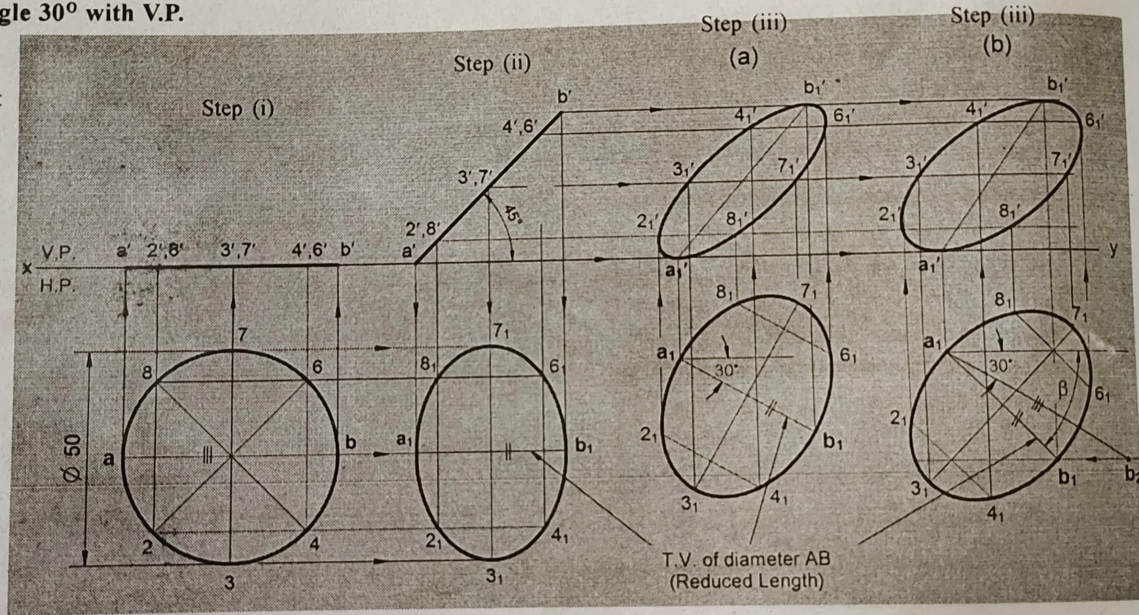


Fig. 6.68